OS Orange Level Problem Report

Questions

Q5.1 — pes checkout <branch>

- **In .pes/:** Update HEAD to point to the new branch, rebuild the index to match the new branch's tree
- **Working directory:** Replace all tracked files with the versions from the new branch
- **Why it's complex:** You have to handle uncommitted changes, create/delete/update files carefully, and avoid corrupting the repo if something fails midway

Q5.2 — Detecting Dirty Working Directory

1. Hash the current working file → compare with the hash stored in the **index** → if different, file is dirty
2. Compare that file's hash in the **current branch vs target branch** → if they differ AND file is dirty → **refuse checkout** (would overwrite user's changes)

Q5.3 — Detached HEAD

- Commits still get created, but no branch pointer moves, so only HEAD tracks them
- Once you switch branches, those commits become **unreachable and lost**
- **Recovery:** Create a branch at that commit hash — git branch recover <hash>. In real Git, reflog helps you find the hash

---

Q6.1 — Garbage Collection Algorithm

1. **Mark:** Start from all branch tips → walk every commit → collect all referenced trees and blobs into a HashSet

2. **Sweep:** Delete any object in .pes/objects/ whose hash isn't in the set

Data structure: HashSet (O(1) lookups)

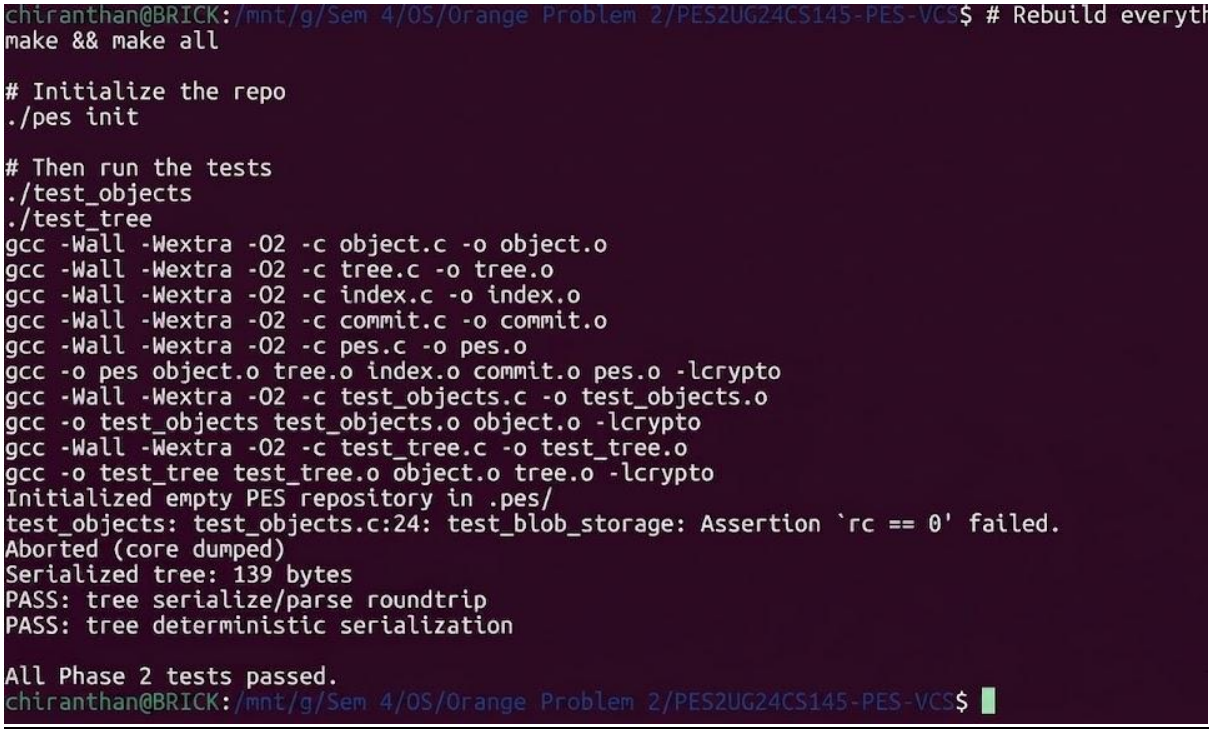
Estimate: ~100k commits + ~100k trees + ~1M blobs = **~1.1 million objects**

Q6.2 — GC Race Condition

Problem: GC scans branches → finds no reference to a new blob → deletes it → meanwhile, a commit finishes writing and now references the deleted blob → **repo corrupted**

How Git avoids it: Never delete objects newer than ~2 weeks old, so in-progress writes are always safe

Screenshots



```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ # Rebuild everyth
make && make all

# Initialize the repo
./pes init

# Then run the tests
./test_objects
./test_tree
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Initialized empty PES repository in .pes/
test_objects: test_objects.c:24: test_blob_storage: Assertion `rc == 0' failed.
Aborted (core dumped)
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ make && make all
gcc -Wall -Wextra -02 -c object.c -o object.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
chiranthan@BRICK:/mnt/g/Sem Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes init
Initialized empty PES repository in .pes/
chiranthan@BRICK:/mnt/g/Sem Orange Problem 2/PES2UG24CS145-PES-VCS$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aaf
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem
/PES2UG24CS145-PES-VC$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7
34b3377afedb9b46a1af8e11ec4f476

.pes/objects/2a/594d39232787fba8eb7287418aec9
8ec8fc2ecdaf5aaf2e650ed471e566fcf

.pes/objects/d5/8213f5dbe0629b5c2fa2e5c7d4213
09227ed0221bbe9ddb5e5c4b9aafc12

chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem
PES22UG24CS145-PES-VCVCS$
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./test_
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization
All Phase 2 tests passed.
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VC$
```

```
hiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ find .pes/objects -type f
pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedbb9ba1461af8e11ec4f476
pes/objects/2a/594d3922787fba8eb72877418eee99cbcc2ecdaf5aafe650eda471e566fcf
pes/objects/d5/8213f5dbe0629b5c2fa28ec7d21ea09227ed0221bbe9b55ec4baafc12
hiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9ba1461af8e11ec4f476 | head
0000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
e content.
```

```
chiranthanK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes init
Initialized empty PES repository in .pes/
chiranthanK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ echo "hello" > file1.txt
chiranthanK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ echo "world" > file2.txt
chiranthanK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes add file1.txt file2
chiranthanK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ cat .pes/i

100777 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 17766876
100777 e00c50e16a2df38f8d6bf809e181ad0)248da6e6719f35f9f7e65d6f606199f7f 6 file1
100777 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e99f7f 1776687662 6 file2
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-$
```

```
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes init
Initialized empty PES repository in .pes/
chiranthan@BRICK:/mSem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ echo "Hello" > hello.txt
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes add hello.txt
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes commit -m "Initial co
Committed: 18ffe4b6b514... Initial commit
chiranthan@BRICK:/mSem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ echo "World" >> hello.txt
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes add hello.txt
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes commit -m "Add world"
Committed: 64648a9d254d... Add world
chiranthan@BRICK:/mSem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ echo "Goodbye" > bye.txt
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes add bye.txt
chiranthan@BRICK:/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes commit -m "Add farewe
Committed: 6a55b77f89cb... Add farewell
chiranthan@BRICK:/mSem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ ./pes log
commit 6a55b77f89cbdede8cd11daf427de641a71bff80d2e10345f7fa7a849a6086d6
Author: Dasari chiranthan <PES2UG24CS145>
Date:   1776695321

    Add farewell

commit 64648a9d254d0d4ca4928b5ca67209011ecc7e51adf861357a8caa922ca63c62
Author: Dasari chiranthan <PES2UG24CS145>
Date:   1776695273

    Add world

Add farewell

commit 18ffe4b6b5141373c29449aea3f12816a5a450ee3a99ffa9500fadb842db59be
Author: Dasari chiranthan <PES2UG24CS145>
Date:   1776695215

    Initial commit iQ

chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VC$
cat .pes/refs/heads/main
6a55b77f89cbdede8cd11daf427de641a71bff80d2e10345f7fa7a849a6086d6
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VC$
ref: refs/heads/main
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VC$
cat .pes/HEAD
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
ref: refs/heads/main
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ find .pes -type f | s
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/18/ffe4b6b5141373c29449aea3f12816a5a450ee3a99ffa9500fadb842db59be
.pes/objects/64/648a9d254d0d4ca4928b5ca67209011ecc7e51adf861357a8caa922ca63c62
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6a/55b77f89cbdede8cd11daf427de641a71bff80d2e10345f7fa7a849a6086d6
.pes/objects/a4/34f8b1bf184edcfb0a2eb35fd6d405ae02fa246b7205d26ad9f5c087266025
.pes/objects/d6/dfaf358f4da1a003e5a16a953e496962f93ecb6809f767d8bbfc3a926a0b58
.pes/objects/e6/7ed66bch4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/ea/abeb710475acf500f6636b1507e573174a5a47d09906b1917edc5fe4fb17fb
.pes/refs/heads/main
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
```

```
    Update file.txt

commit 947db5c38e3e76533535ade2f175c1a0cfbbbbc13be78f4e8d656399e054edf1
Author: Dasari Vinay <PES2UG24CS145>
Date:   1776695803

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
0040d7d0cdc23da4698a9f96ff3ecf2ae6f1b4c8f4eadbf85085c23e656a6bf5

--- Object Store ---
Objects created:
10
.pes/objects/00/40d7d0cdc23da4698a9f96ff3ecf2ae6f1b4c8f4eadbf85085c23e656a6bf5
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/94/7db5c38e3e76533535ade2f175c1a0cfbbbbc13be78f4e8d656399e054edf1
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/cc/ab193274743744883978d1b35924475bae5bc76bf668a5de3c4a643168c99c
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ make test-integrat
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
test_sequence.sh: line 26: /mnt/g/Sem: No such file or directory
make: *** [Makefile:45: test-integration] Error 127
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$
```

```
chiranthan@BRICK:/mnt/g/Sem 4/OS/Orange Problem 2/PES2UG24CS145-PES-VCS$ make test-int
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 947db5c38e3e... Initial commit

Log after first commit:
commit 947db5c38e3e76533535ade2f175c1a0cfbbbbc13be78f4e8d656399e054edf1
Author: Dasari Chiranthan <PES2UG24CS145>
Date:   1776695803

    Initial commit


--- Second Commit ---
Committed: ccab19327474... Update file.txt

--- Third Commit ---
Committed: 0040d7d0cdc2... Add farewell

--- Full History ---
commit 0040d7d0cdc23da4698a9f96ff3ecf2ae6f1b4c8f4eadbf85085c23e656a6bf5
Author: Dasari Chiranthan <PES2UG24CS145>
Date:   1776695803

    Add farewell

commit ccab19327474374488397 8d1b35924475bae5bc76bf668a5de3c4a643168c99c
Author: Dasari Chiranthan <PES2UG24CS145>
Date:   1776695803
```